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ECEN 3730 – PCB Design Manufacture
10/24/25

Board 2 Report

Objective:

The objective of designing Board 2 is to demonstrate how PCB layout choices directly affect switching noise and understanding good design and measurements practices. These techniques are implemented on our two-layer board. The board must feature two hex-inverter circuits. One with good practices like a continuous return plane and a close decoupling capacitor while the other utilizes intentionally bad practices, such as long ground routing and a far decoupling capacitor. The measurement differences are compared.

Project Overview:

1. The circuit must be designed correctly in Altium passing all Design Rule Checks.
2. Must include a 5V to 3.3V LDO
3. Must include a 555-timer producing a square output of approximately 500Hz with an approximate 66% duty cycle.
4. Must include a switch between 5V and 3.3V LDO output to each hex
5. Must include a switch between to send the 555 timer's output to a chosen hex
6. Must include test points for the hex output as well as its quiet high and quiet low
7. Must have one side of the board utilize bad design practices (far decoupling capacitor and a lack of a ground plane)
8. All components must be soldered neatly and correctly as good practice.

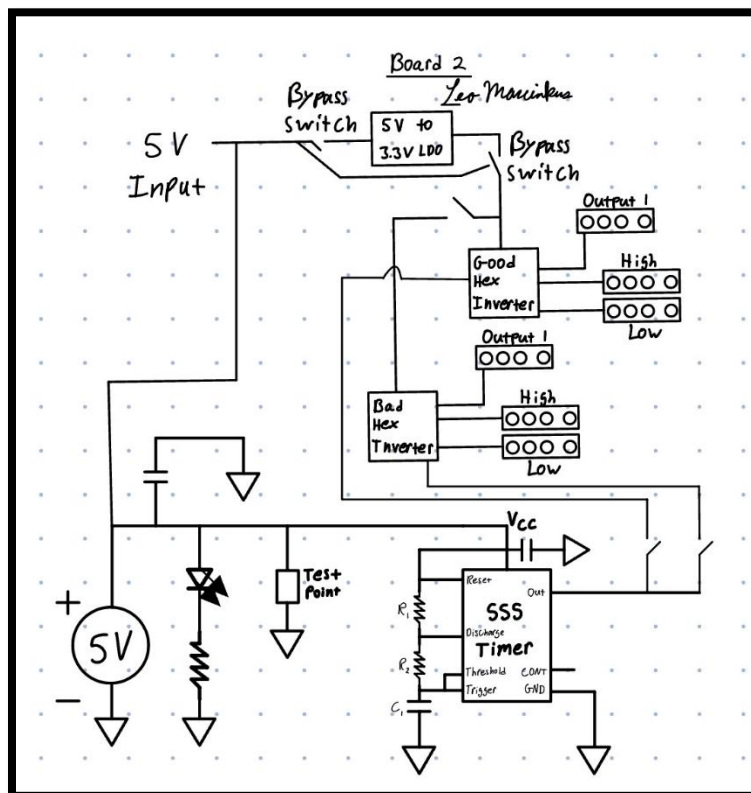


Figure 1: Napkin Sketch

Schematic and PCB:

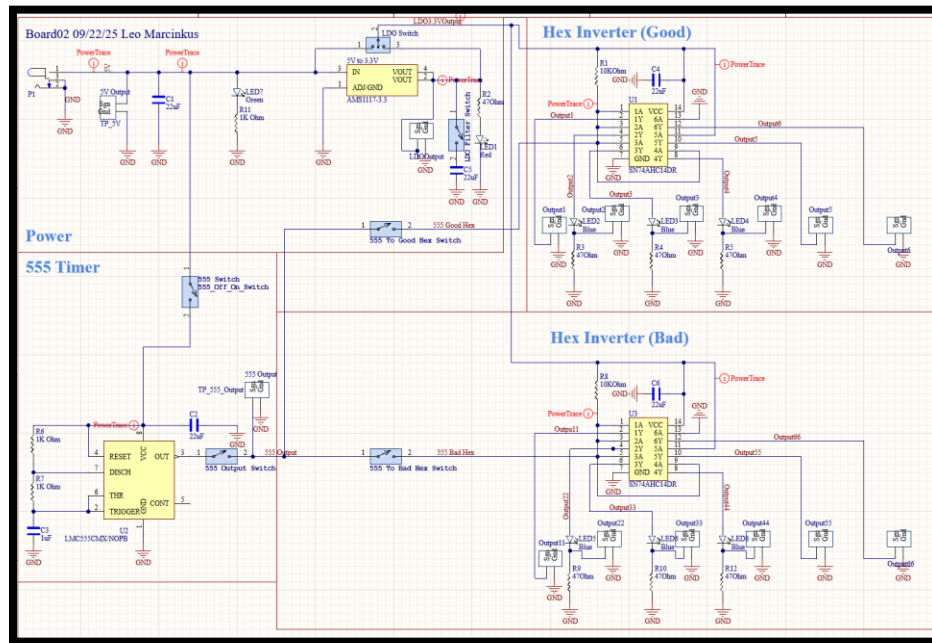


Figure 2: Altium Schematic

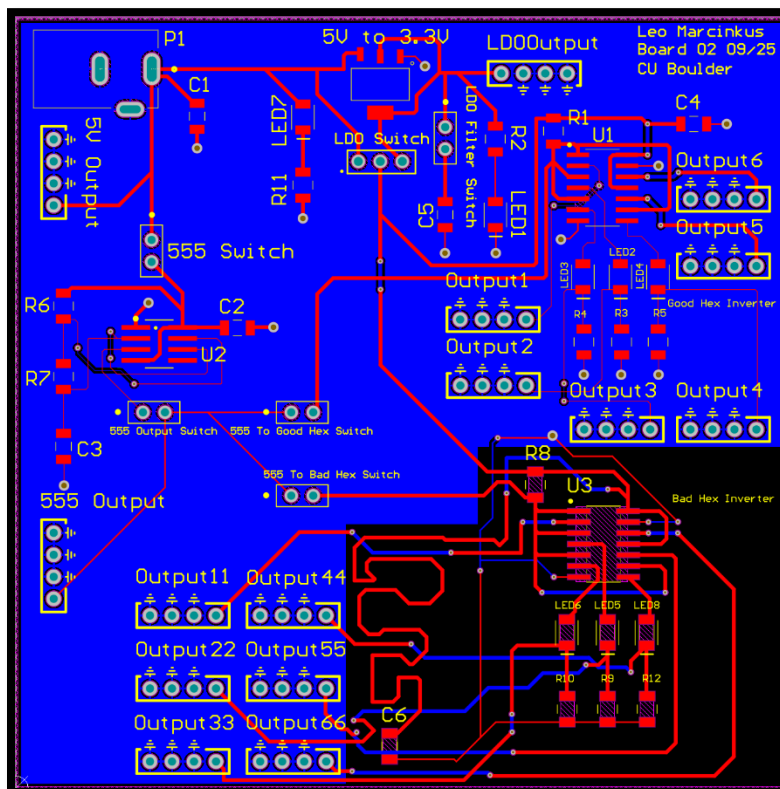


Figure 3: Altium PCB

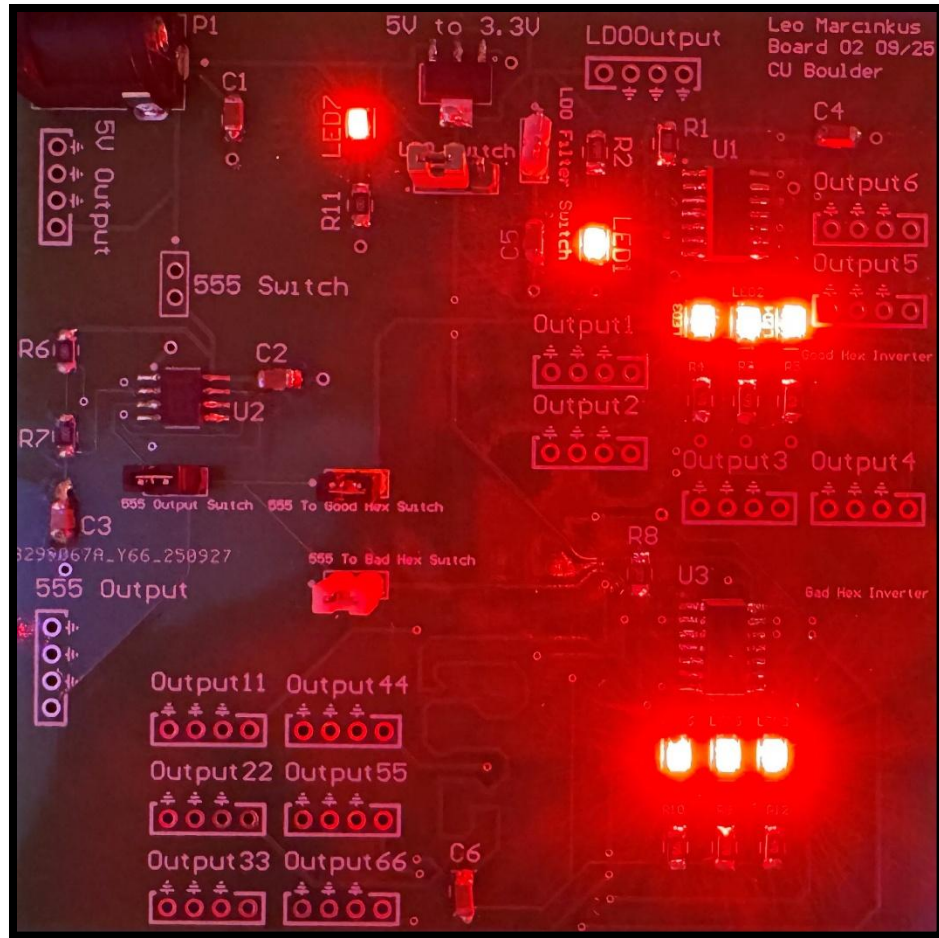


Figure 4: Completed and Powered Board 2

Oscilloscope Measurements:

Good Hex Inverter at 3.3V:

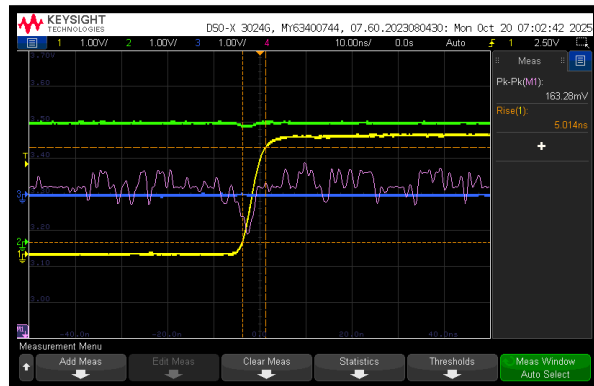


Figure 5: Rise Time and Rail Collapse of Good Hex at 3.3V

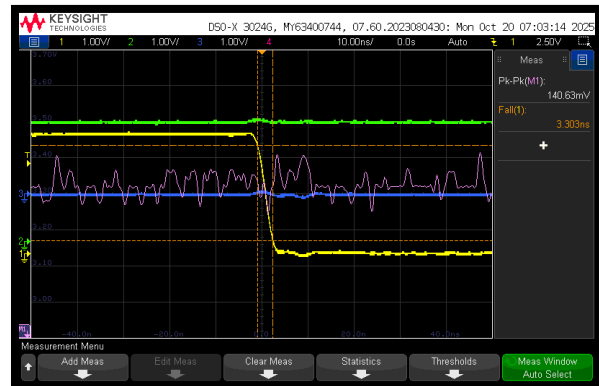


Figure 6: Fall Time and Rail Collapse of Good Hex at 3.3V

Bad Hex Inverter at 3.3V:

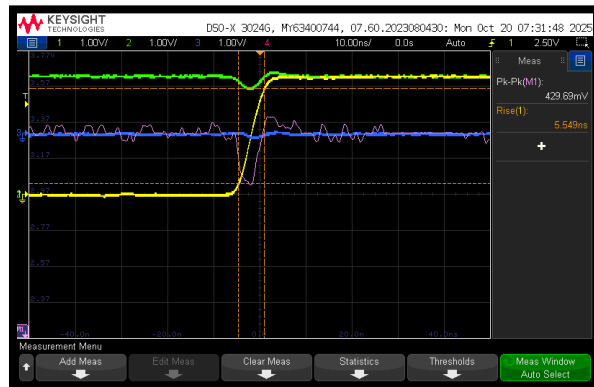


Figure 7: Rise Time and Rail Collapse of Bad Hex at 3.3V

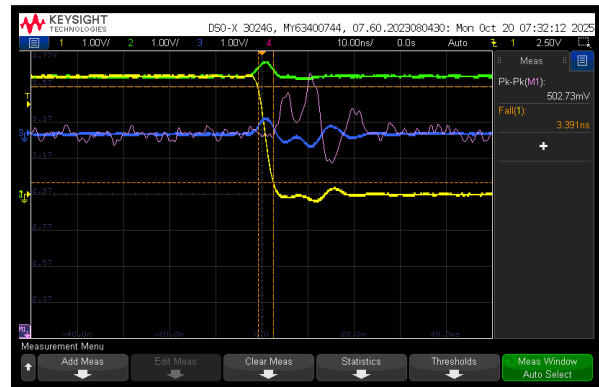


Figure 8: Fall Time and Rail Collapse of Bad Hex at 3.3V

Good Hex Inverter at 5V:

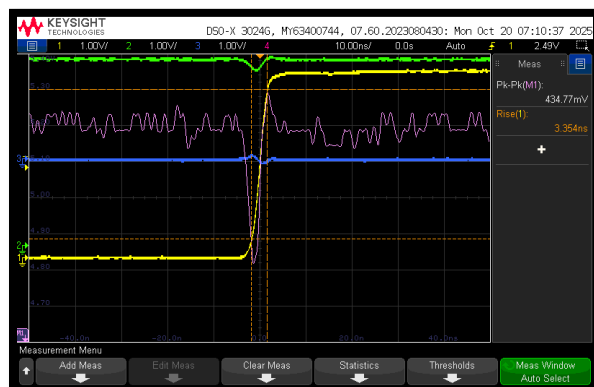


Figure 9: Rise Time and Rail Collapse of Good Hex at 5V

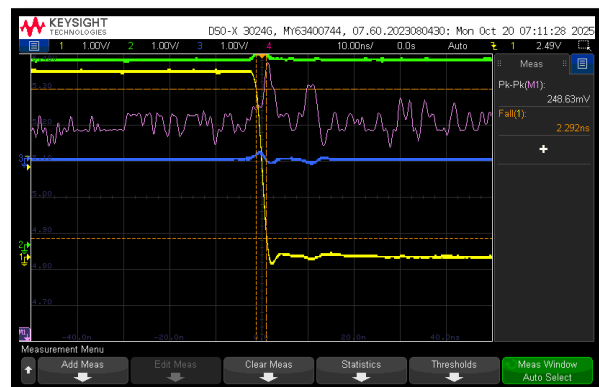


Figure 10: Fall Time and Rail Collapse of Good Hex at 5V

Bad Hex Inverter at 5V:

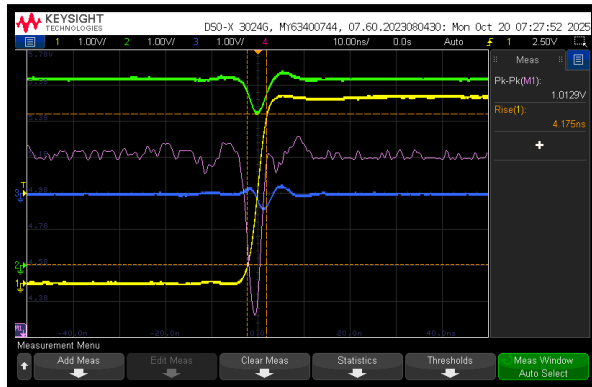


Figure 11: Rise Time and Rail Collapse of Bad Hex at 5V

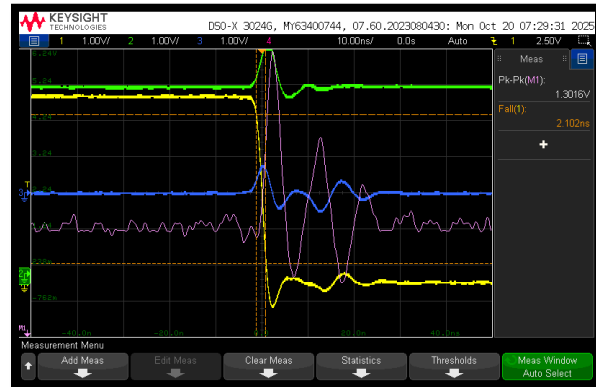


Figure 12: Fall Time and Rail Collapse of Bad Hex at 5V

Results:

The rise and fall time of each hex inverter's output is measured. The voltage rail collapse is determined by utilizing the oscilloscope's math function. This is done by subtracting the quiet high and quiet low outputs of the hex.

Hex Inverter	Rise Time	Fall Time	Voltage Rail Collapse (At Rise)
Good at 3.3V	5.014 ns	3.303 ns	163.28 mv
Bad at 3.3V	5.549 ns	3.391 ns	429.69 mv
Good at 5V	3.354 ns	2.292 ns	434.77 mv
Bad at 5V	4.175 ns	2.102 ns	1.0129 V

Switching noise of the power rail is also measured.

Good Hex Inverter at 3.3V Switching Noise:

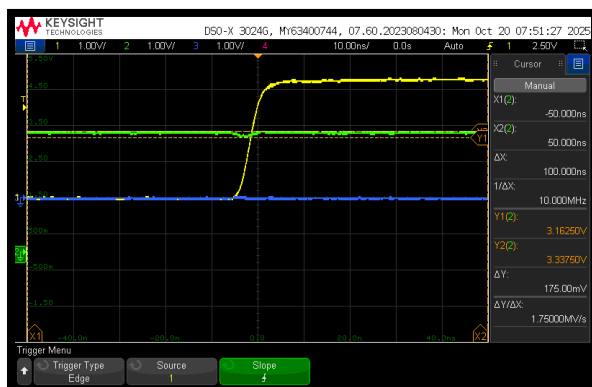


Figure 13: Switching noise at rising edge of Good Hex at 3.3V

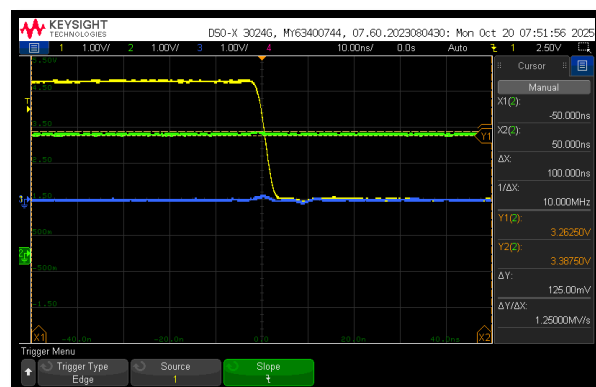


Figure 14: Switching noise at falling edge of Good Hex at 3.3V

Bad Hex Inverter at 3.3V Switching Noise:

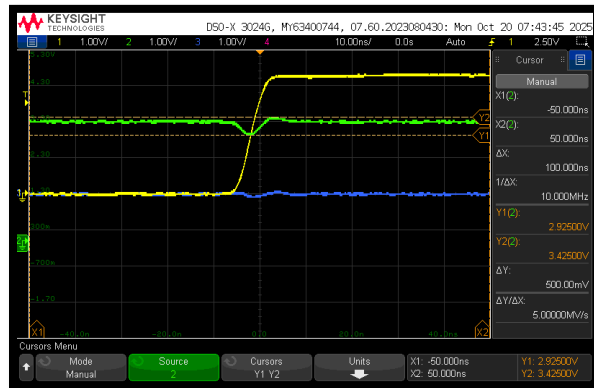


Figure 15: Switching noise at rising edge of Bad Hex at 3.3V

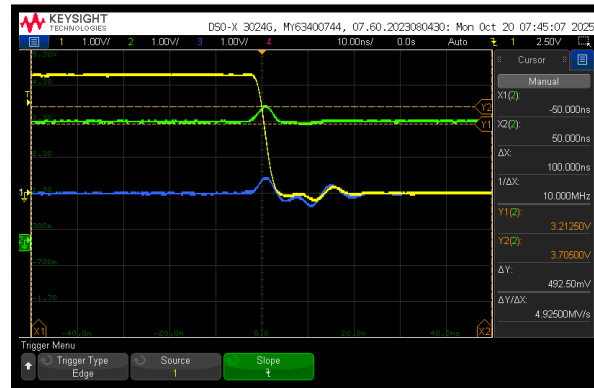


Figure 16: Switching noise at falling edge of Bad Hex at 3.3V

Good Hex Inverter at 5V Switching Noise:

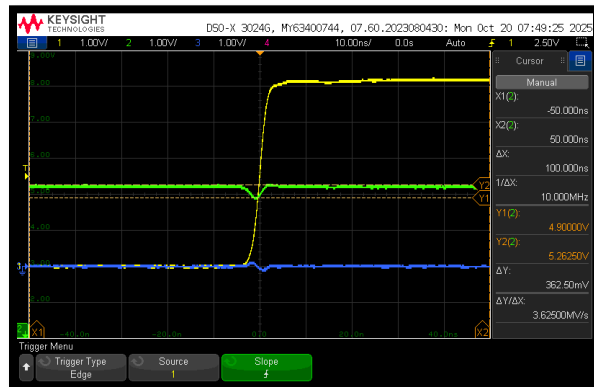


Figure 17: Switching noise at rising edge of Good Hex at 5V

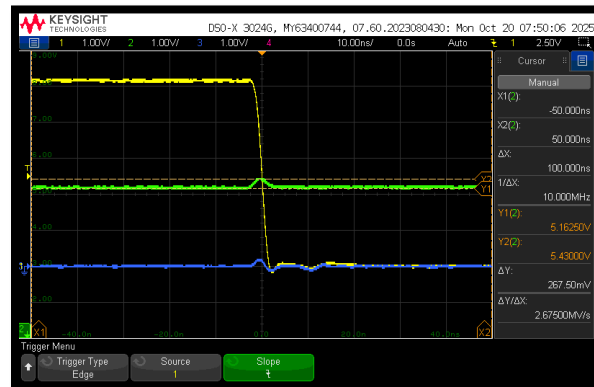


Figure 18: Switching noise at falling edge of Good Hex at 5V

Bad Hex Inverter at 5V Switching Noise:

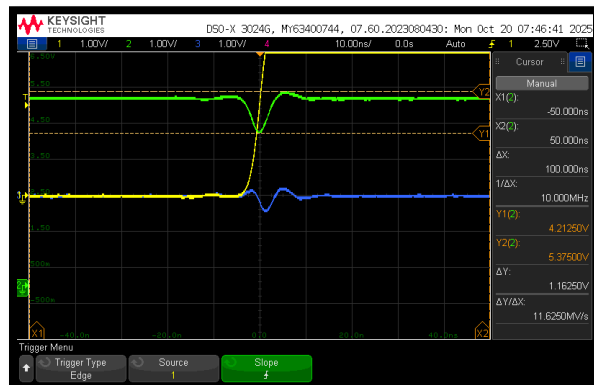


Figure 19: Switching noise at rising edge of Bad Hex at 5V

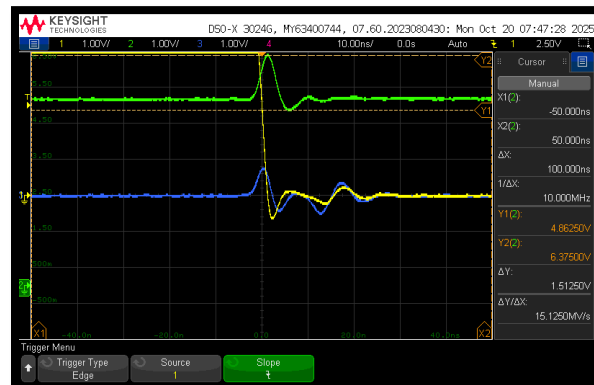


Figure 20: Switching noise at falling edge of Bad Hex at 5V

Results:

The switching noise of each hex inverter's output is measured. This is done by measuring the noise of the high rail during the rise and fall of the hex's output.

Hex Inverter	Switching Noise (Rise)	Fall Time (Fall)
Good at 3.3V	175.00 mV	125.00 mV
Bad at 3.3V	500.00 mV	492.50 mV
Good at 5V	362.50 mV	267.50 mV
Bad at 5V	1.16250 V	1.51250 V

Lastly, the output oscillations of the LDO were measured with and without its 22 μ F capacitor filter.

3.3V LDO on without filtering capacitor:

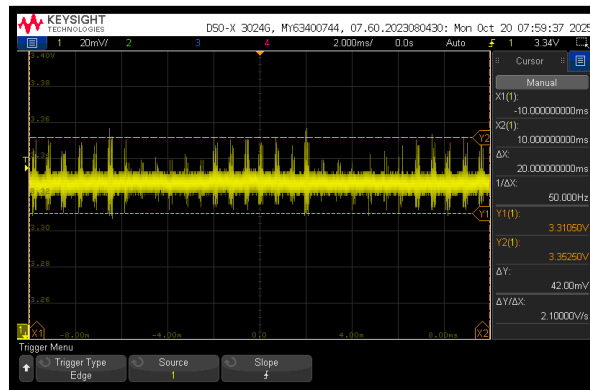


Figure 21: 3.3V on without filter

3.3V LDO on with filtering capacitor:

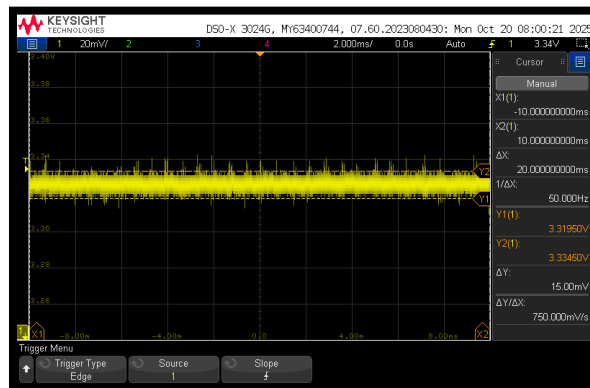


Figure 22: 3.3V on with filter

Bypass 5V on without filtering capacitor:

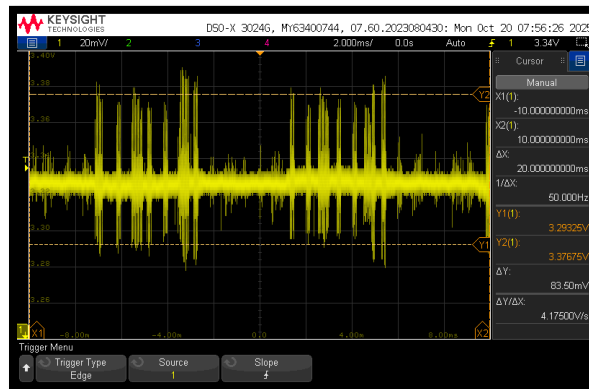


Figure 23: 5V on without filter

Bypass 5V on with filtering capacitor:

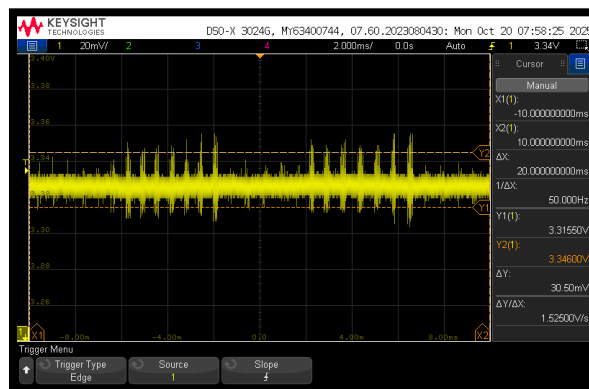


Figure 24: 5V on with filter

Results:

The noise of each voltage output is measured.

Voltage	Noise
3.3V out without filter	42.00 mV
3.3V out with filter	15.00 mV
5V out without filter	83.50 mV
5V out with filter	30.50 mV

Measurement Analysis:

There is a noticeable difference in performance between the hex inverter when running at 3.3V vs 5V. There is an even more noticeable difference in performance in the good practices vs bad practices designs of the hex inverters.

3.3V:

- The rise time for the 3.3V good practices hex inverter is 1.1067 times better than the 3.3V bad practices hex inverter
- The fall time for the 3.3V good practices hex inverter is 1.0266 times better than the 3.3V bad practices hex inverter
- The rail collapse for the 3.3V good practices hex inverter is 2.6316 times better than the 3.3V bad practices hex inverter

5V:

- The rise time for the 5V good practices hex inverter is 1.2248 times better than the 5V bad practices hex inverter
- The fall time for the 5V good practices hex inverter is 1.0904 times **worse** than the 5V bad practices hex inverter
- The rail collapse for the 5V good practices hex inverter is 2.3297 times better than the 5V bad practices hex inverter

Good practices 3.3V vs good practices 5V:

- The rise time for the 3.3V good practices hex inverter is 1.4948 times **worse** than the 5V bad practices hex inverter
- The fall time for the 3.3V good practices hex inverter is 1.441 times **worse** than the 5V bad practices hex inverter
- The rail collapse for the 3.3V good practices hex inverter is 2.6627 times better than the 5V bad practices hex inverter

3.3V Switching Noise:

- The rise time switching noise for the 3.3V good practices hex inverter is 2.8571 times better than the 3.3V bad practices hex inverter
- The fall time switching noise for the 3.3V good practices hex inverter is 3.9400 times better than the 3.3V bad practices hex inverter

5V Switching Noise:

- The rise time switching noise for the 5V good practices hex inverter is 3.2069 times better than the 5V bad practices hex inverter
- The fall time switching noise for the 5V good practices hex inverter is 5.6542 times better than the 5V bad practices hex inverter

LDO Filter Capacitor:

- The 3.3V LDO with the filter capacitor is 2.800 times better than without the filter capacitor
- The 5V output with the filter capacitor is 2.7377 times better than with the filter capacitor

What Worked:

1. The board functions entirely. Each hex inverter provides a stable output with the ability to measure its quiet high and quiet low as well
2. The 5V to 3.3V LDO functions properly. The switch to choose which voltage gets sent to the hex inverter works as intended
3. The 555 timer implemented from the last board still works
4. Measurements were taken for rise times, fall times, rail collapse, and switching noise. All values that were measured appear to be within reason compared to the board measured in Lab 15

Analysis of Project:

1. *What worked well and should be carried over to the next board?*
 - a. Clear labels made assembly of the board easy as it was simple to tell what component should go where. Separating sections of the board significantly organized the design of the circuit. Including a power indication LED helped visualize what was happening after assembly. Having multiple isolation switches in the design helped with measurements being able to fully isolate part of the circuit.
2. *What should be done differently?*
 - a. Although there were no issues, I think better component placement during the design phase would be beneficial. Some components during assembly were difficult to solder if another component was too close. I also added way to many test points which took up valuable real estate on the PCB.
3. *Any hard errors?*
 - a. No hard errors occurred preventing function.
4. *Any soft errors that can be corrected for the next board?*
 - a. No specific errors occurred when recording data. But including every component's value on the overlay of the board would be useful for the soldering process.

Conclusion:

Board 2 was successfully designed and assembled. The board utilizes a 5V powered 555-timer to output to a good and bad hex inverter. Three probes were used to measure the output of each hex, and its quiet high and quiet low. These measurements are used to calculate the rail collapse. The switching noise and LDO oscillations with and without a filter were investigated as well. Measurements were taken to understand the behavior of this circuit and how decoupling capacitor location as well as a ground plane can affect results. Overall, I this board taught me good circuit design practices while also improving my soldering skills. Through the analysis of this board, the next design will feature multiple minor improvements such as including a very close decoupling capacitor.